Energy In Waves

> **SYNOPSIS**
>
> We round up the notes with equations for calculating energy in waves.

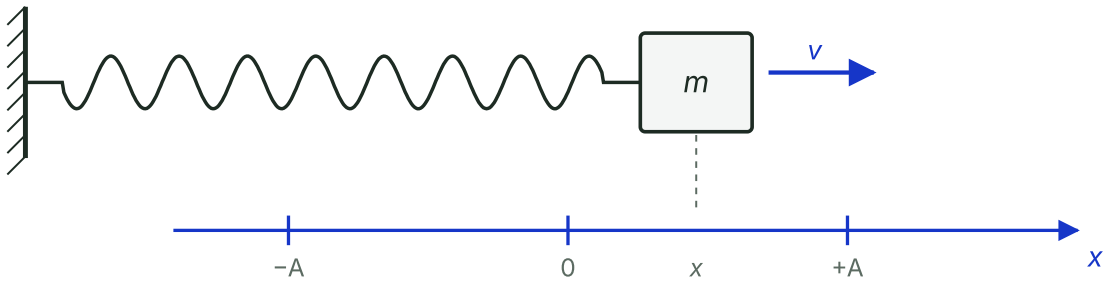
BEFORE YOU READ ON — PREDICT

Pull the block back to $x = A$ and let go. **(1)** Where in the swing is it moving fastest? **(2)** At half the amplitude, $x = A/2$, is it moving at half its top speed? **(3)** If you pull it back twice as far, does the top speed double? Write down your three answers. One formula at the end of this note settles all three.

PERFORMANCE EXPECTATIONS

1. **Write** the total energy of an oscillator as $\frac{1}{2}kA^2$ and say why it does not depend on time.
2. **Trace** the exchange between kinetic and potential energy through one full cycle.
3. **Obtain** the speed at any displacement, $v(x) = \omega\sqrt{(A^2 - x^2)}$.
4. **Reach** $\omega = \sqrt{(k/m)}$ from energy conservation alone, and say how that route differs from the one taken earlier.
5. **State** where in the swing the speed is greatest and where it is zero, and justify it from the energy.

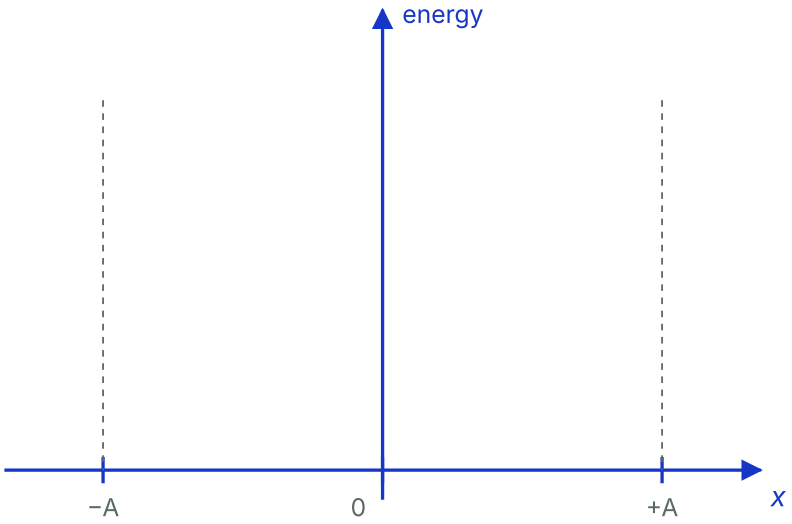
2. Energy in waves (just SHM)



The block on the spring from Note 4, caught somewhere between the turning points. It swings between $x = -A$ and $x = +A$, and at the instant drawn it is at x, moving.

Nothing is lost to friction here, so whatever energy you put in by pulling the block back is still there at every later instant. It just moves between two accounts: the spring's and the block's.

1. Write the total energy of the block at a general position x, in terms of m, v, k and x.
2. At $x = \pm A$ the block is momentarily at rest. What is E there?
3. At $x = 0$ the spring is relaxed. What is E there?
4. Those are the same E. Set them equal and get v_{max}.
5. From Note 4 you know v_{max} a second way, in terms of ω and A. Write it, equate the two, and read off ω. Then write f and T.



Plot the three energies against position here. Mark the value of each at $x = 0$ and at $x = \pm A$, and say which one is flat and why.

THE PUNCHLINE

The angular frequency comes out of the spring and the mass, and **nothing else**. Not the amplitude, not how you started it, not where it is in the swing. Pull it back twice as far and it still takes exactly as long to come back — which is why a pendulum clock keeps time as its swing dies away, and why ω is worth writing in a box.

v at any position

The last step gave the speed at one point of the swing. The same conservation statement gives it everywhere, because the energy at the turning point is the energy at every other point too.

6. Write "the energy at $x = \pm A$ equals the energy at a general x", and solve it for v.
7. Rewrite that with ω in place of $\sqrt{(k/m)}$. Check it at $x = 0$ and at $x = \pm A$.

Looking ahead

You now have the energy an oscillator carries and how fast it is moving at every point of its swing. **Note 11** asks what happens when that energy *leaves*: how fast it goes out, how it thins as it spreads, and how we end up measuring it in decibels.

Where this sits. Note 4 got $\omega = \sqrt{(k/m)}$ by writing Newton's second law for the block and solving the equation of motion that came out. This note gets the identical result from a single sentence about energy, with no calculus at all. Both are worth having: the first says *why* the motion is sinusoidal, the second says what is conserved while it happens.

BACK TO YOUR PREDICTION

(1) Fastest at the centre, $x = 0$. That is the one point where the spring holds none of the energy, so the block holds all of it. **(2) No — and this is the one worth being wrong about.** At $x = A/2$, $v = \omega\sqrt{(A^2 - A^2/4)} = (\sqrt{3}/2)\omega A \approx$ **87%** of the top speed. Energy goes as x^2, so at half the displacement only a quarter of it has gone into the spring and three quarters is still motion. **(3) Yes, exactly double.** $v_{max} = \omega A$ is linear in A, and ω does not depend on A at all — so twice the pull gives twice the speed in the same period.